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Paleomagnetic results from the Late Carboniferous/Early Permian Casper Formation: implications for northern Appalachian tectonics

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Paleomagnetic samples were collected from 190 m of the Late Carboniferous/Early Permian Casper Formation in southeastern Wyoming. A total of 549 samples was drilled near the vicinity of Horse Creek Station at an average stratigraphic interval of 33 cm. All samples were reversely magnetized. Rock magnetic analyses indicate that the primary carrier of remanence in the formation is hematite. A selection criterion applied to the partial demagnetized data restricted the sample population to 233, resulting in a paleomagnetic North Pole located at 47.4°N, 127.4°E ($\delta_p = 0.7$; $\delta_m = 1.4$). The Casper pole agrees well with other Late Carboniferous/Early Permian poles for cratonic North America. The tight clustering of these paleomagnetic poles suggests that little apparent polar motion with respect to North America occurred during this time. Comparing the stable North American poles with paleomagnetic poles from Late Carboniferous/Early Permian strata of the New England–Canadian Maritime region (Acadia) indicates that this region did not reach its present position relative to North America until at least the Early Permian.

1. Introduction

Kent and Opdyke [1,2] have shown that systematic differences between Devonian and Early Carboniferous paleomagnetic poles from the coastal New England–Canadian Maritime region and cratonic North America could be explained if the New England–Maritime region was located 1500 km southward of its present position during the Late Devonian/Early Carboniferous. Van der Voo et al. [3] and Irving [4] have also presented data which support this hypothesis. Exactly when the New England–Maritime region or "Acadia" [2] reached its present position with respect to cratonic North America is less well defined. Based on similarities between paleomagnetic poles of Late Carboniferous age from the two areas, Kent and Opdyke [1] concluded that the New England–Maritime region reached its present position by the Late Carboniferous. However, as Kent and Opdyke admit, their analysis was limited, in that virtually all Late Carboniferous paleomagnetic poles for North America come from the New England–Maritime region. Their only "reliable" paleomagnetic pole of Late Carboniferous age was that of Christensen and Helsley [5]—a preliminary pole for the Upper Carboniferous portion of the Maroon Formation of Colorado. Clearly, additional paleomagnetic data for Late Paleozoic rocks are needed to determine exactly when this terrain

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became attached to stable North America.

We have sampled the Casper Formation of southeastern Wyoming in order to define better the Late Carboniferous/Early Permian paleomagnetic pole for cratonic North America. It would now appear that the New England–Maritime region did not reach its present position relative to North America until at least the Early Permian.

2. Geology

The Casper Formation though locally eolian [6] is predominantly a shallow-water marine sequence of interbedded sandstones and carbonates. The formation crops out mainly along the flanks of the Laramie Range of southeastern Wyoming (Fig. 1), ranging in thickness from greater than 300 m on the east side to less than 50 m on the west side. It is underlain by and intercalated with the Pennsylvanian Fountain Formation [8]. The Fountain Formation rests unconformably upon rocks of Precambrian age. The Casper is conformably overlain by red siltstones and shales of the Permian Opeche (Stanka) Shale [8].

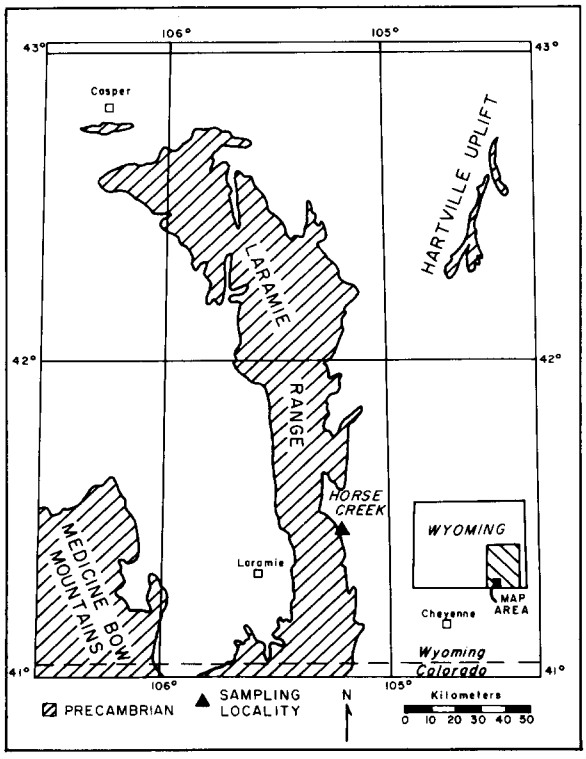


Fig. 1. Index map of southeastern Wyoming showing the location of the sampling area (modified from Thompson et al. [7]).

The units of the Casper Formation show considerable lithologic variation. The sandstones are reddish-brown to pink to yellowish-brown in color, fine-grained, well-sorted, highly crossbedded to thin bedded or massive, subarkosic to orthoquartzitic. The carbonates are gray, tan, pink, purple, lavendar or white, finely crystalline, dense massive, thin bedded, cherty, fossiliferous limestones, dolomitic limestones, and dolomites.

The age of the Casper Formation has been established as Desmoinesian through Wolfcampian on the basis of brachiopod, fusulinid and conodont studies [9–14]. Agatston [11] has divided the Casper Formation into three divisions (lower, middle, upper) based on age and lithic characteristics. The lower division of the Casper Formation is Desmoinesian in age as indicated by forms of *Fusulina* and *Mesolobus*. The lower division consists of arkose, sandstone, shale, and limestone in order of importance. The middle division of the Casper Formation is a thick carbonate sequence separated by thin sandstone interbeds. Fusulinids (forms of *Triticites*) from the middle division indicate that this unit belongs to the Missourian and Virgilian Stages of the Late Pennsylvanian. The upper division of the Casper Formation is Wolfcampian and consists of dolomitic limestones, dolomites and sandstones. Possible local disconformities exist between both the lower-middle and middle-upper divisions of the Casper Formation.

3. Locations and sampling

Sampling of the Casper Formation was undertaken at two locations near the vicinity of Horse Creek Station (41.45°N, 105.21°W) on the east side of Laramie Range (Fig. 1). Here the Casper Formation consists entirely of marine sediments [6] that form spectacular hogbacks flanking the Laramie Range. These strata represent a simple drape on the Precambrian (Laramie Range) basement which is non-plunging (Blackstone, personal

communication, 1979). The first sampling site was located on North Horse Creek (41.46°N, 105.22°W) where the lower half of the Casper Formation is exposed along the flanks of the stream cut. The other sampling site was located one mile south of the Horse Creek site in Baker Canyon (41.45°N, 105.20°W) where the upper half of the Casper crops out along road and stream cuts. At both sampling locations the sandstone and limestone units show a consistent strike of N19°W and with a dip of 81° to the east. A distinctive purple crinoid limestone traceable between the two sites provided lithostratigraphic control. The total thickness of the Casper at these two locations is approximately 220 m, of which 190 m were sampled.

Samples were collected using a hand-held gasoline-powered drill and oriented "in situ" using a magnetic compass. At each locality, samples were collected at measured stratigraphic intervals of approximately 33 cm. In all, 549 samples from 190 m of the Casper Formation were collected.

4. Treatment

In the laboratory, samples were cut into as many 2.5-cm-long cylinders as possible (usually two). Natural remanent magnetization (NRM) measurements were made on a ScT cryogenic magnetometer. Samples were thermally demagnetized in a horizontal, non-inductively wound furnace of 24 sample capacity housed in six nested mu-metal cans, and cooled in three nested mu-metal cylinders which reduced the field to less than 15 gammas. Samples were then stored in two nested mu-metal cans where the field was less than 50 gammas until they were measured.

5. NRM and thermal demagnetization results

The NRM directions of all samples from the Casper Formation are shown in Fig. 2a, uncorrected for geological dip, and in Fig. 2b, corrected for geological dip. Fig. 3 shows a histogram of the NRM intensity (J_n) for these sandstone and carbonate samples. The geometric mean value of

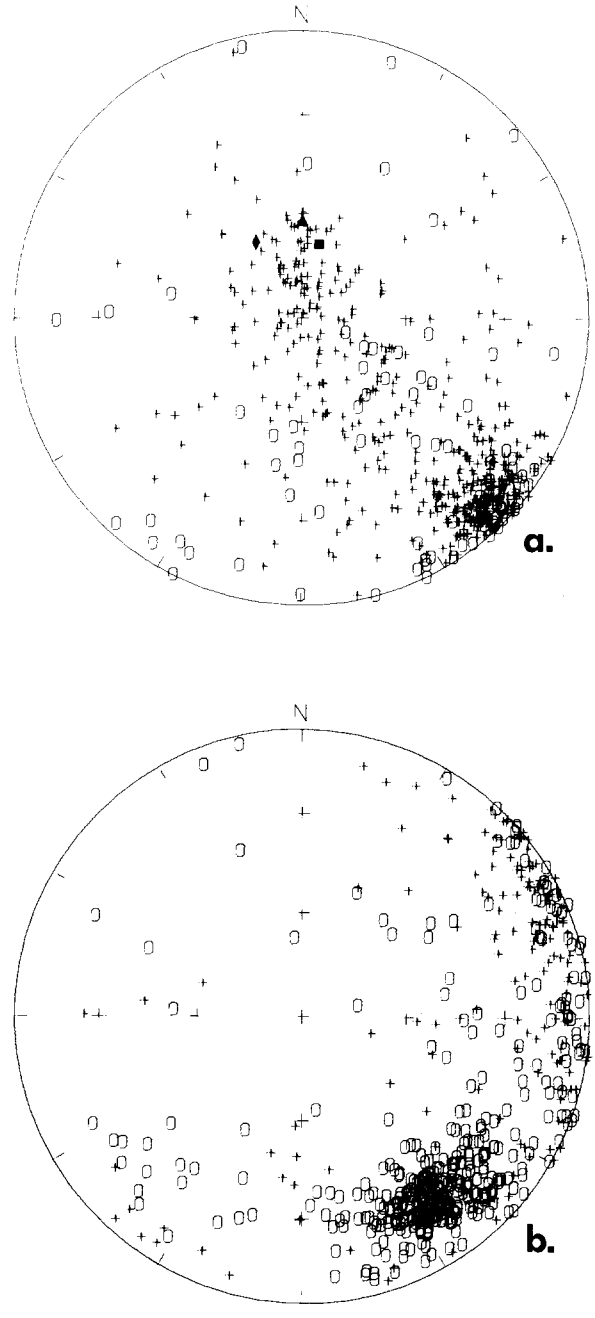


Fig. 2. NRM directions (N=549) of the Casper Formation: (a) before structural correction and (b) after structural correction (strike=341°, dip=81°E). Projection is equal area. Open symbols are in the upper hemisphere, + in the lower hemisphere. The present-day field direction (■), the geocentric axial dipole direction (▲), and the expected Cretaceous direction (♦), for the sampling area are also shown.

J_n for the sandstone samples ($N = 250$) is 1.28×10^{-3} A/m while the geometric mean for the carbonate samples ($N = 299$) is 5.41×10^{-4} A/m.

Several features are apparent from Fig. 2a,b. First, much of the data (40–50%) after structural correction (Fig. 2b) show a characteristic Late Paleozoic direction (shallow upward inclination toward the southeast)—an indication of magnetic stability with time for these samples. Secondly, many of the sample directions (Fig. 2a) lie in a streak between the uncorrected Late Paleozoic cluster and a secondary direction which appears to be a mixture of the present day ($D = 13°$; $I = 68.5°$), the present geocentric axial dipole ($D = 0°$; $I = 62°$), and the expected Cretaceous ($D = 330°$; $I = 64°$) direction for this area. Some random scatter also exists. These directions are from samples that had J_n's below 5×10^{-5} A/m.

Some sample directions in Fig. 2a exhibit steep upward inclinations to the south-southeast—very reminiscent of a Late Cretaceous/early Tertiary reversed direction that would be expected for this area. Some of these directions represent scatter in the data but the more strongly magnetized samples (21 in all) are located in discrete zones within the upper carbonate units of the middle division of the formation. These zones (5 in all) vary from 1.2 to 2.5 m in thickness and are lithologically very similar to other carbonate samples collected stratigraphically above and below these zones. It seems highly unlikely that these directions could be the result of short period irregularities (excursions) of the Permo-Carboniferous field since studies [15] indicate that the field was quite stable during this time interval. Since these directions are very similar to a Late Cretaceous or early Tertiary direction that would be expected for this area, it would appear that these samples have acquired a secondary magnetization during or after folding of the Casper Formation while the geomagnetic field was of reversed polarity. Exactly why these samples are found in discrete zones is unknown. They may be related to changes in the groundwater regime or some other secondary event associated with the folding of the sediments.

A group of samples were selected from the different sandstone and carbonate units for Curie temperature analysis, isothermal remanent magnetization (IRM) acquisition curves, and thermal demagnetization experiments. These samples represented most of the different lithologic units encountered at the sampling localities.

A recording thermomagnetic balance similar to those of Schwarz [16] and Doell and Cox [17] was used to determine the Curie temperature of the magnetic phases present in some of the samples. All samples were heated and cooled in air. Curie temperature analyses indicate that hematite was the primary carrier of remanence in both the sandstones and carbonates. Typical magnetization (J_s)

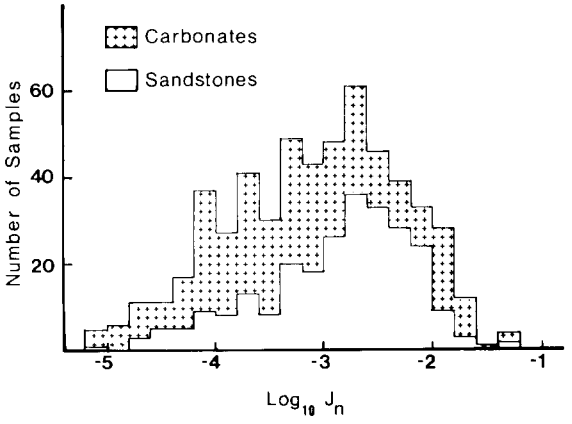


Fig. 3. Histogram of NRM intensities, J_n, for the sandstone and carbonate samples of the Casper Formation.

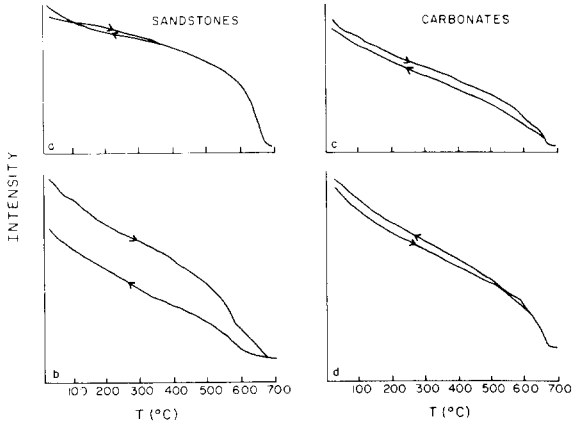


Fig. 4. Representative sandstone and carbonate thermomagnetic curves.

curves exhibiting a hematite Curie point for both the sandstones and carbonates are shown in Fig. 4. The initial concave upward appearance upon heating of these curves is the probable result of a large component of paramagnetic minerals in the samples [18]. The small inflections seen in the curves of Fig. 4b,c near the magnetite Curie point suggest that a small amount of magnetite is present in these samples. The irreversible nature of the curve in Fig. 4b suggests possible oxidation of magnetite to hematite during heating. The hematite occurs mainly as two forms in the sandstones—red pigment and a black oxide mineral. The red pigment occurs as a stain around the grains and as cement in the interstices of the sandstone [19]. The black oxide mineral was identified as specularite by use of a binocular microscope on a magnetic separate of a disaggregated sample. In the carbonates, hematite occurs as red pigment and in the red clays incorporated in the limestones [20].

Successive IRM's were given to other samples in progressively increasing fields (up to 1.0 T) using a 4-inch water-cooled electromagnet. The IRM acquisition curves from the sandstone and carbonate samples show two prominent features (Fig. 5a). First, the IRM curves are concave upward near the origin. If a significant amount of magnetite were present in these samples, the IRM curves would initially be convex upward near the origin even if the magnetite were only 0.5–1.0% of the hematite content in the sample [21]. Therefore, both the IRM acquisition curves and the Curie point curves indicate that hematite is the predominant magnetic mineral in these samples. Secondly, most of the curves show no tendency to level off in fields up to 1.0 T. This behavior can be attributed to a hematite in pigment form which saturates in fields in excess of 1.0 T. In addition to pigment carrying part of the remanence in these samples, the coercivity spectrum (Fig. 5b) of some of the samples indicate the presence of specular hematite (specularite) and possibly a minor amount of magnetite [21].

Thermal demagnetization curves (Fig. 6) also indicated that in most samples hematite has a broad range of blocking temperatures. This is in contrast to many studies of red sediments [22–25] where the hematite had a narrow range of blocking

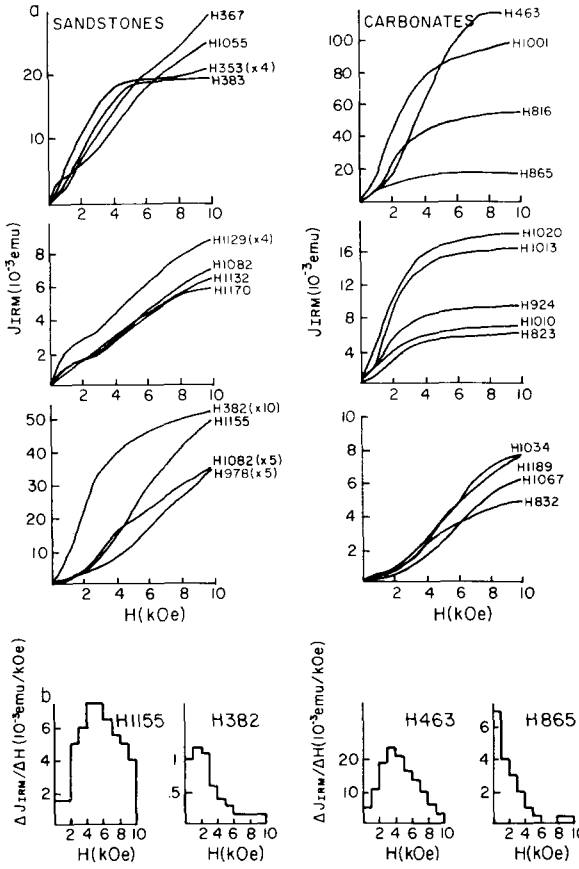


Fig. 5. Isothermal remanence curves (a) and coercivity spectra (b) for representative samples from the various sandstone and carbonate units. Acquisition curves based on data collected at 0.3, 0.6, 1.0, 2.0, 3.0, 4.0, 6.0, 8.0, 10.0 kOe fields (0.02, 0.06, 0.1, 0.2, 0.3, 0.4, 0.6, 0.8, and 1 tesla field).

temperatures near its Curie point. Collinson [26], in studying the importance of pigment and specularite in the remanent magnetization of red sediments, has demonstrated that in samples where hematite has a narrow range of blocking temperatures, specularite is the dominant carrier of remanence whereas in samples where hematite has a broad range of blocking temperatures, pigment of varying grain sizes [18] is the dominant carrier of remanence in the sample. In a study of the Bloomsburg red beds, Irving and Opdyke [22] reported similar conclusions. Therefore, the dominant carrier of remanence in the sediments of the

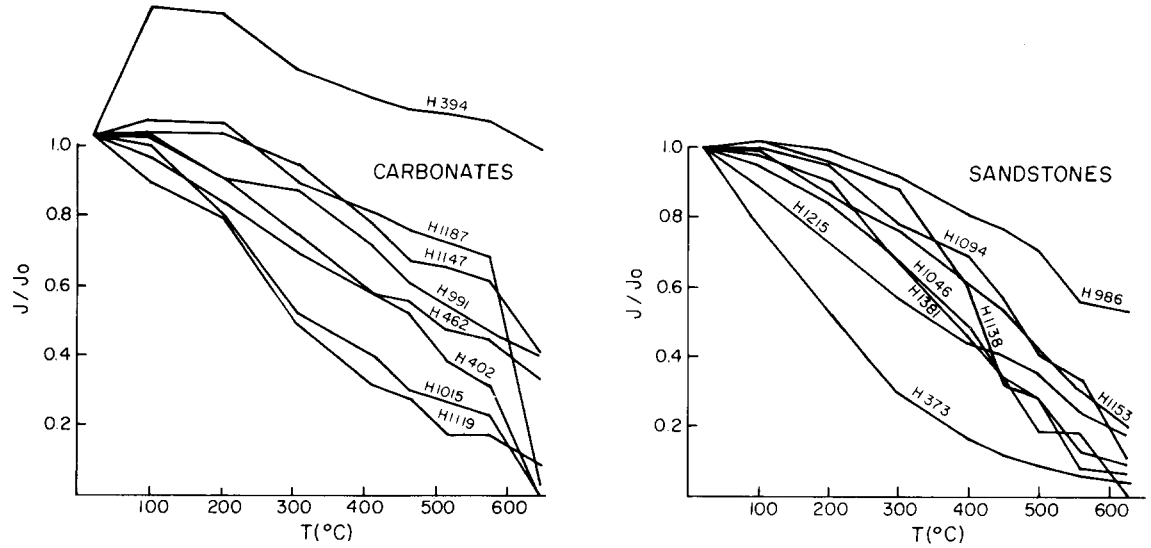


Fig. 6. Normalized thermal demagnetization curves for representative samples from the different sandstone and carbonate units.

Casper Formation appears to be hematite in pigment form.

Vector endpoint diagrams [27] showing the change in direction and intensity of the NRM with increasing temperature of thermal demagnetization for some of the representative pilot samples are shown in Fig. 7. These samples as well as all other pilot samples were demagnetized at temperatures of 100, 200, 300, 400, 450, 500, 560, and 630°C. All vector endpoints in these diagrams have been plotted uncorrected for structural tilt of the formation.

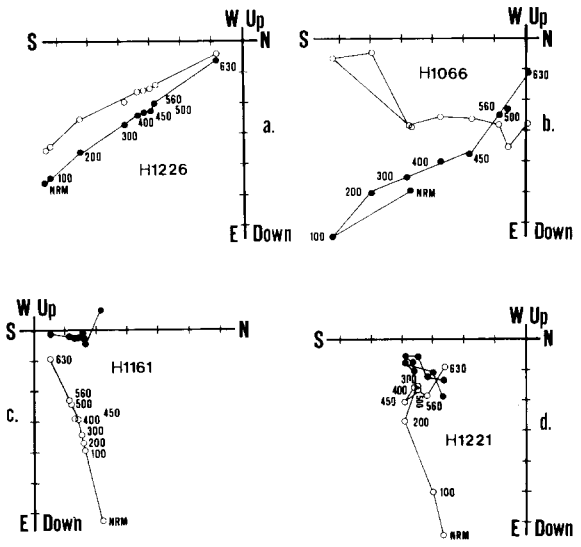


Fig. 7. Vector and point diagrams of Casper Formation samples during thermal demagnetization. Solid circles are projections on the horizontal plane; open circles are on the north-south vertical plane. Thermal demagnetization levels given in degrees Celsius. Horizontal and vertical axes divisions are: (a) 1.2×10^{-3} A/m; (b) 0.36×10^{-4} A/m; (c) 0.40×10^{-3} A/m; and (d) 0.86×10^{-4} A/m.

Samples whose NRM showed a characteristic Late Paleozoic direction exhibited two distinct types of behavior:

(1) A large portion of these samples had vector endpoints that decayed linearly to the origin with increasing temperature (Fig. 7a). Present-day overprints when present were minimal.

(2) Some samples, however, upon heating responded with an initial change in direction and, in many cases, with an increase in intensity (Fig. 7b). The vector removed by this low-temperature demagnetization generally pointed to the north and down and is interpreted to be a present-day overprint. With increasing temperatures of demagnetization these samples displayed roughly linear trends that pass by the origin. In many instances the vector being removed is directed to the southeast with a shallow inclination—characteristic of a Late Paleozoic direction. Therefore, it would appear that a secondary component of magnetization

exists in these samples whose blocking temperature spectrum exceeds that of the characteristic Late Paleozoic component.

Samples whose NRM indicated a secondary direction of magnetization can also be divided into two basic groups based on their response to heating. In the first group (Fig. 7c), much like the first group above, a single component of magnetization could be isolated. However, in many of these samples the removal of a low-temperature component first occurred, followed by linear decay of the vector endpoints to the origin. The vector defined by this linear decay to the origin points approximately to the north and steeply down and appears to be either a present-day, an axial dipole, or a Cretaceous direction. This single-component behavior of a vector that is clearly secondary in origin suggests that in many of the Casper samples the characteristic magnetization has been completely replaced by a secondary magnetization. The other group of samples having a secondary NRM direction had trajectories that were very erratic with heating (Fig. 7d). At higher temperatures a change in direction was often accompanied by an increase in intensity of the sample. In some instances roughly linear trends could be picked out over three or four heating steps, but these did not trend toward the origin. Vectors that were removed varied from sample to sample, but most either directed to the north and down or to the southeast and near horizontal or steeply upward. The magnetization in these samples appears to be comprised of at least two, if not three, components whose blocking temperature spectra completely overlap. Therefore, isolation of a meaningful direction of magnetization in these samples was not possible even with higher temperatures of demagnetization.

Samples from the different sandstone units were thermally demagnetized at 450, 560, and 630°C and in some cases, 650°C. Samples from the carbonate units were demagnetized at temperatures of 300, 450, and 530°C. Many of the carbonate samples that were weakly magnetized to begin with (see Fig. 3) by 530°C had intensities below the noise level of the magnetometer.

In spite of secondary magnetic overprints, polarity information could be obtained from all samples by looking at their sequential directional change with increasing thermal demagnetization. If the sample's directional change was toward the Late Paleozoic reversed direction, then the sample was considered reversely magnetized. No samples showed a sequential movement toward a Late Paleozoic normal direction. At any one level of thermal demagnetization, less than 1% of all the samples measured had northwesterly directions (270–360°). These samples were at least twenty-five arc degrees from being antipodal to the mean of the stably magnetized reversed samples of this study, and show erratic directional behavior from one thermal demagnetization level to another. Secondary magnetization acquired during the Tertiary may have camouflaged the presence of a normal polarity zone, but this seems unlikely since no samples showed any indication of containing a stable normal component of magnetization. Therefore, all samples from this study are considered reversely magnetized.

6. Analysis of remanent directions

As previously discussed, many of the samples from this study have NRM directions that group near or lie in a streak between a secondary (present day/axial dipole/Cretaceous) field direction and a Late Paleozoic direction. Helsley [28] has shown that the closer the NRM direction lies to the secondary field direction, the greater the probability that the thermally demagnetized direction will contain a mixture of both characteristic and secondary magnetic components. Use of such sample directions could yield erroneous estimates for pole position calculations. Therefore, any sample whose direction is a mixture of both characteristic and secondary components should be eliminated from any data set where the primary goal of the study (e.g., this study) is to define a paleomagnetic pole position.

To determine which samples had a thermally demagnetized direction that contained a significant secondary component of magnetization, the demagnetized results for each sample were plotted on a orthogonal vector diagram and analyzed according to the technique of Zijderveld [27]. If

the demagnetized results did not define a linear trend toward the origin, the sample was rejected on the premise that its magnetization was probably multicomponent. Therefore, samples in which an unchanging single component of magnetization could not be isolated by thermal demagnetization were eliminated from the sample population. For samples that did demonstrate single component behavior, the directions that defined each linear trend were vectorially averaged to determine the mean direction for that sample. The thermally demagnetized directions from these single component samples were generally no more than 10° from their undemagnetized NRM direction.

It may be argued that a linear trend may be too hard to evaluate with only three of four data points. However, it was our philosophy that if a linear trend to the origin was not easily recognizable from the vector endpoints, that sample should not be used in the pole position calculation. Examples of easily recognizable linear trends to the origin can be seen in Fig. 8. Some samples that were eliminated probably would not have been if other reliability criteria had been employed. Addition or deletion of these samples in a large data set of well clustered points such as ours did not significantly alter the final results.

The above selection criteria reduced the sample population to 263 or 48% of the thermally demagnetized samples. Characteristic directions from these samples are shown in Fig. 9, corrected for geological dip. The calculated mean for these samples is declination, 143.4° and inclination, −19.6° ($\alpha_{95} = 3.2°$; kappa = 8.2) with an angular dispersion (δ) about the mean of 28.4°. As seen in Fig. 9, some sample directions diverge significantly from the cluster of directions in the southeast quadrant. When uncorrected for structural tilt, these directions either group near a present day/axial dipole/Cretaceous field direction or near a Late Cretaceous-early Tertiary reversed direction. These directions (30 in all) are clearly secondary and are more than two standard deviations (>57°) from the mean of the group. By

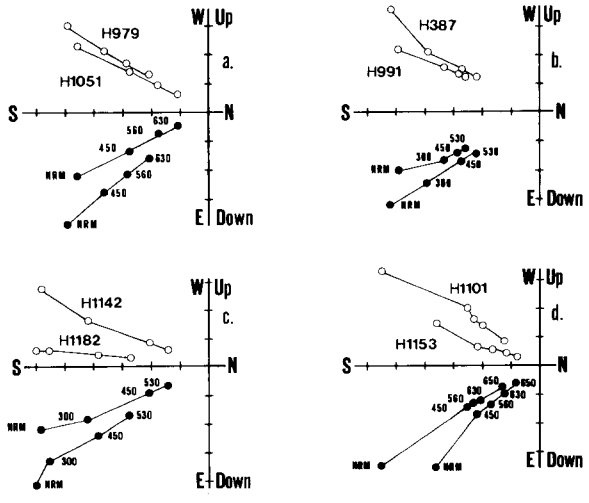


Fig. 8. Orthogonal vector diagrams from samples of the Casper Formation showing a linear decay to the origin. Vectors are shown after correction for structural tilt. Solid circles are projections on the horizontal plane; open circles are on the north-south vertical plane. Horizontal and vertical axes divisions are: (a) 1.4×10^{-3} A/m; (b) 0.68×10^{-3} A/m; (c) 0.2×10^{-3} A/m; (d) 0.80×10^{-3} A/m.

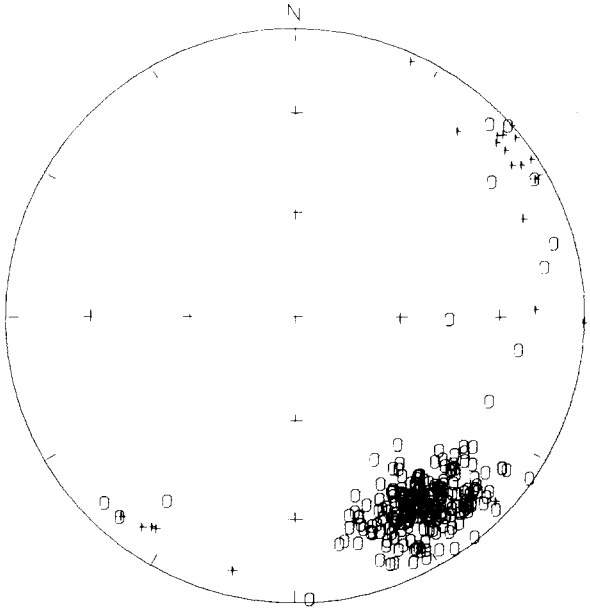


Fig. 9. Characteristic directions of magnetization ($N=263$) for the thermally demagnetized samples of the Casper Formation that showed single component behavior upon thermal demagnetization. Data are structurally corrected (strike=341°; dip=81°E). Open symbols are in the upper hemisphere, +in the lower hemisphere. Equal-area projection.

TABLE 1

Summary of paleomagnetic data for the Casper Formation

Stratigraphic age	D (°)	I (°)	N	R	K	α_{95} (°)	δ	North Pole location	δ_p, δ_m
Late Carboniferous/ Early Permian	146.9	−20.0	233	228.15	47.9	1.4	11.7	47.4°N, 127.4°E	0.7, 1.4
Pennsylvanian	145.0	−19.2	161	156.99	39.86	1.8	12.8	45.9°N, 129.1°E	1.0, 1.9
Early Permian	150.9	−21.8	72	71.45	129.3	1.5	7.1	50.6°N, 123.4°E	0.8, 1.6

analogy with planar distributions, there is only a 5% probability or less that these directions actually belong to the group, therefore, they were eliminated and the mean direction for the formation recalculated. Table 1 summarizes the pertinent paleomagnetic data. The mean direction after correction for tilt of the strata is now declination, 146.9° and inclination, −20.0°; this represents a change of only 3.3° from the previously calculated mean. The resulting paleomagnetic pole is located at 47.4°N, 127.4°E ($\delta_p = 0.7°$; $\delta_m = 1.4°$); the small oval of confidence results from the large number of samples. Arranging these samples stratigraphically, a progressive change in direction of magnetization (declination—more southerly; inclination—more negative) is observed upward through the formation. Dividing these samples according to stratigraphic age gives a mean Pennsylvanian pole that lies at 45.9°N, 129.1°E ($N = 161$; $\delta_p = 1.0$; $\delta_m = 1.9$) while the Early Permian pole is located at 50.6°N, 123.4°E ($N = 72$; $\delta_p = 0.8$, $\delta_m = 1.6$). (This is a revision of the Pennsylvanian and Early Permian poles reported for the Casper Formation in Diehl and Shive [29].) The relative position of these poles suggests that a small amount of apparent polar motion to the northeast with respect to North America did occur during the remanence acquisition of the Casper strata—a result consistent with the motion indicated by the path of apparent polar wander for cratonic North America for the Late Carboniferous/Early Permian [4,30].

7. Time of magnetization

The age of the characteristic magnetization of the Casper Formation is definitely pre-Laramide folding. The mean direction for the formation prior to structural correction for tilt is declination, 138.7° and inclination, 9.9°, which is definitely not a Late Cretaceous or an early Tertiary direction. Therefore, this magnetization must have been acquired before folding. Consequently, no fold test was performed to determine a pre- or post-folding origin for this magnetization. In fact, the time of acquisition of this stable characteristic magnetization is certainly pre-Triassic. If this were not the case, normal polarity zones should be present in the formation since the Triassic was a time of both normal and reversed polarity in contrast to the Permo-Carboniferous when the geomagnetic field was of reversed polarity. However, since the carrier of the remanence in the formation is a pigmentary hematite, the possibility exists that the magnetization was acquired over a long period of time [31]. Therefore, we cannot entirely rule out a Permian magnetization for the entire formation. Nevertheless, the progressive change in the direction of magnetization upward through the section suggests to us that the time period over which the sediments acquired their magnetization was short compared to the amount of time required for several degrees of apparent polar wander, and for this reason we believe that the magnetization is much closer to being Permo-Carboniferous than Late Permian.

8. Discussion

A compilation of Late Carboniferous/Early Permian paleomagnetic poles for North America is given in Table 2 and shown in Fig. 10. Several observations are rather apparent:

TABLE 2

Late Carboniferous/Early Permian paleomagnetic poles, North America

Unit	Age	Pole (°N/°E)	Reference *
North American Craton			
1. Ingelside Fm.	EP	46/122	Diehl and Shive [29]
2. Casper Fm.	LC-EP	47/127	this study
3. Culter Fm.	EP	44/119	13.50, 13.51
4. Elephant Canyon	EP	44/120	13.52
5. Halgaito Tongue	EP	49/117	11.67
6. Maroon Fm.	LC	41/123	Christensen [32]
7. Dunkard Fm.	LC	44/122	8.88
New England–Canadian Maritime Province			
1. Hurley Creek Fm.	LC	39/125	9.98
2. Prince Edward Island red beds	LC-EP	40/126	8.95
3. Prince Edward Island red beds	LC-EP	42/133	8.99
4. Bonaventure Fm.	LC-EP	38/133	8.100
5. Cumberland Group	LC	36/125	11.76
6. Pictou Group ss.	LC	41/132	8.101
North American Craton average, $N=7$		45/121	$\alpha_{95}=2.6$
New England–Canadian Maritime Province average, $N=6$		39/129	$\alpha_{95}=3.2$

* Numbers refer to McElhinny's lists in the *Geophysical Journal* 1968–1978.

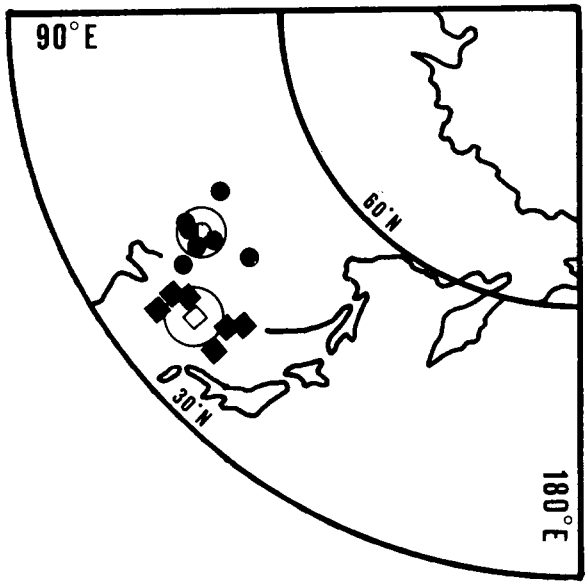


Fig. 10. Late Carboniferous/Early Permian paleomagnetic North Poles for cratonic North America (solid circles) and the New England–Maritime region (solid squares), including their respective means (open symbols) and circles of 95% confidence. Projection is equal-area.

(1) The mean pole position determined from the Casper Formation (127.4°E, 47.4°N) is very similar to other Late Carboniferous/Early Permian paleomagnetic poles from the stable interior of North America.

(2) The tight clustering of these paleomagnetic poles as well as those for the New England–Maritime region indicate that during the Late Carboniferous/Early Permian, little apparent polar motion relative to either region took place.

(3) There is a small but significant difference ($8 \pm 6°$ at the 95% probability level) between paleomagnetic poles from cratonic North America (mean North Pole = 45.0°N, 121.0°E, $\alpha_{95} = 2.6$) and the New England –Maritime region (mean North Pole = 39.4°N, 128.9°E, $\alpha_{95} = 3.2$). This difference may be somewhat artificial since the mean cratonic pole includes more Early Permian studies than the New England–Maritime result. However, we believe that this is not the case since the Late Carboniferous/Early Permian appears to be a time of little apparent polar motion. Therefore, we believe that these results indicate that the

two regions did not share a common apparent polar-wander path at this time.

Kent and Opdyke [1,2] have shown that differences between Devonian and Early Carboniferous poles from cratonic North America and the New England–Maritime area could be explained in terms of relative motion between the two. Their analysis of the available paleomagnetic data indicated that the sinistral motion necessary to bring the New England–Maritime region to its present position with respect to cratonic North America occurred between Devonian and latest Carboniferous times. Our study of the Casper Formation when combined with other Late Carboniferous/Early Permian poles from the stable interior now suggest that this relative motion may not have ceased until at least the Early Permian. However, more data are still needed. Regardless of whether the New England–Maritime region reached its present position relative to stable North America by latest Carboniferous or early Permian times, the magnitude of the difference between the Late Carboniferous/Early Permian poles versus that for Devonian (20°) and Early Carboniferous (14°) poles for the two areas indicates that most of the relative motion was complete by the end of the Carboniferous.

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